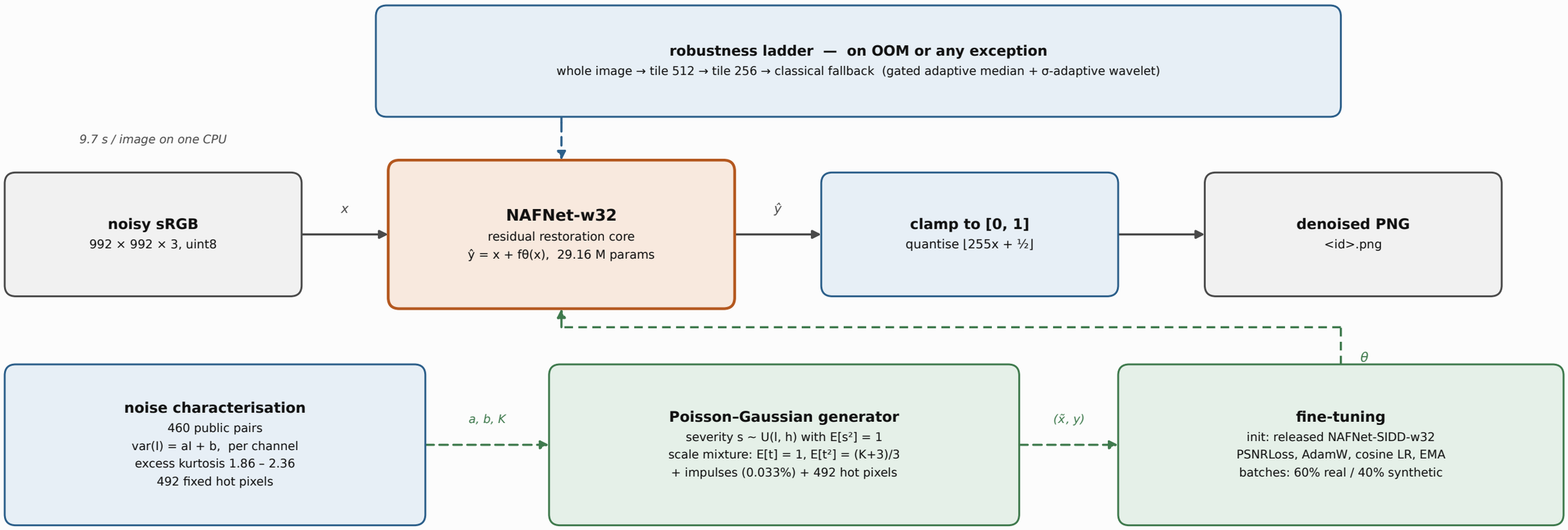
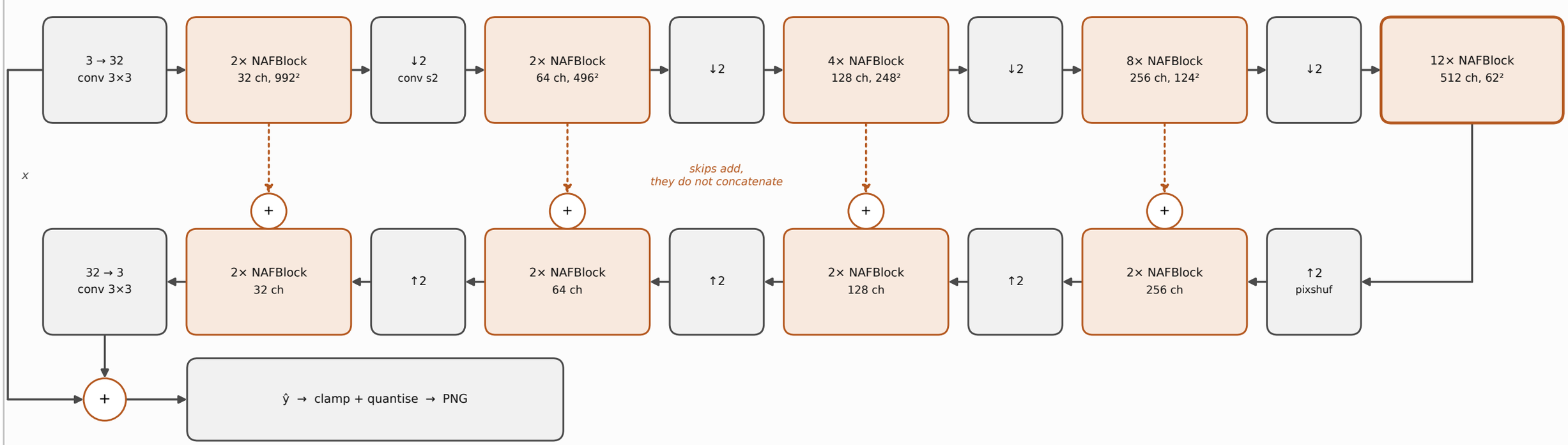
(a) System: measured classical model (blue) → learned core (orange); green runs at training time only



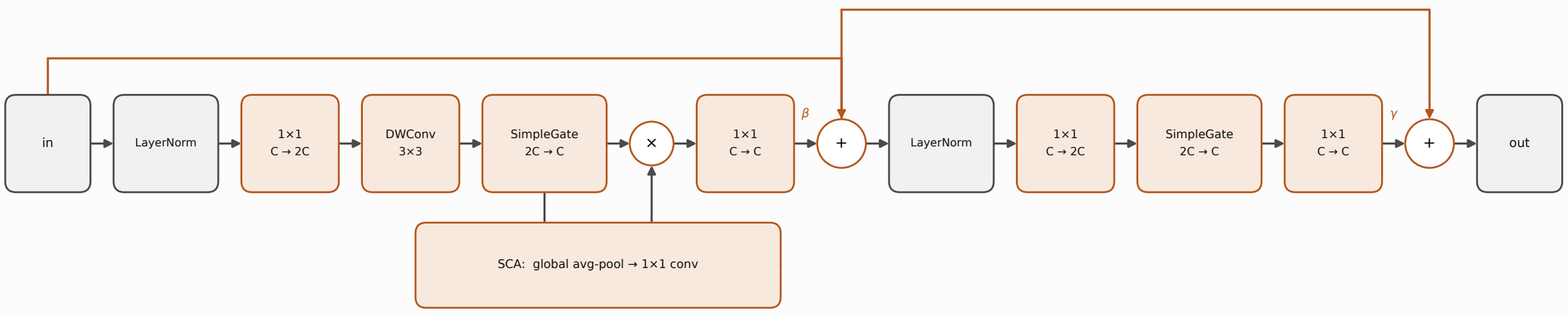
Offline: the generator and the optimiser see the 414 training ids only; the 46 validation ids are held out end to end.

(b) Restoration core: four-level NAFNet U-Net, additive skips, no padding at 992²



992 = $2^5 \cdot 31$, so a four-level U-Net needs zero padding; peak working set 1.28 GB, so the whole image runs in a single pass and tiling is only a fallback.

(c) NAFBlock: activation-free, two residual branches (C channels in, C out)



No ReLU or GELU anywhere: SimpleGate ($x_1 \odot x_2$ on a channel split) and simplified channel attention replace them — the reason the core stays affordable on CPU.